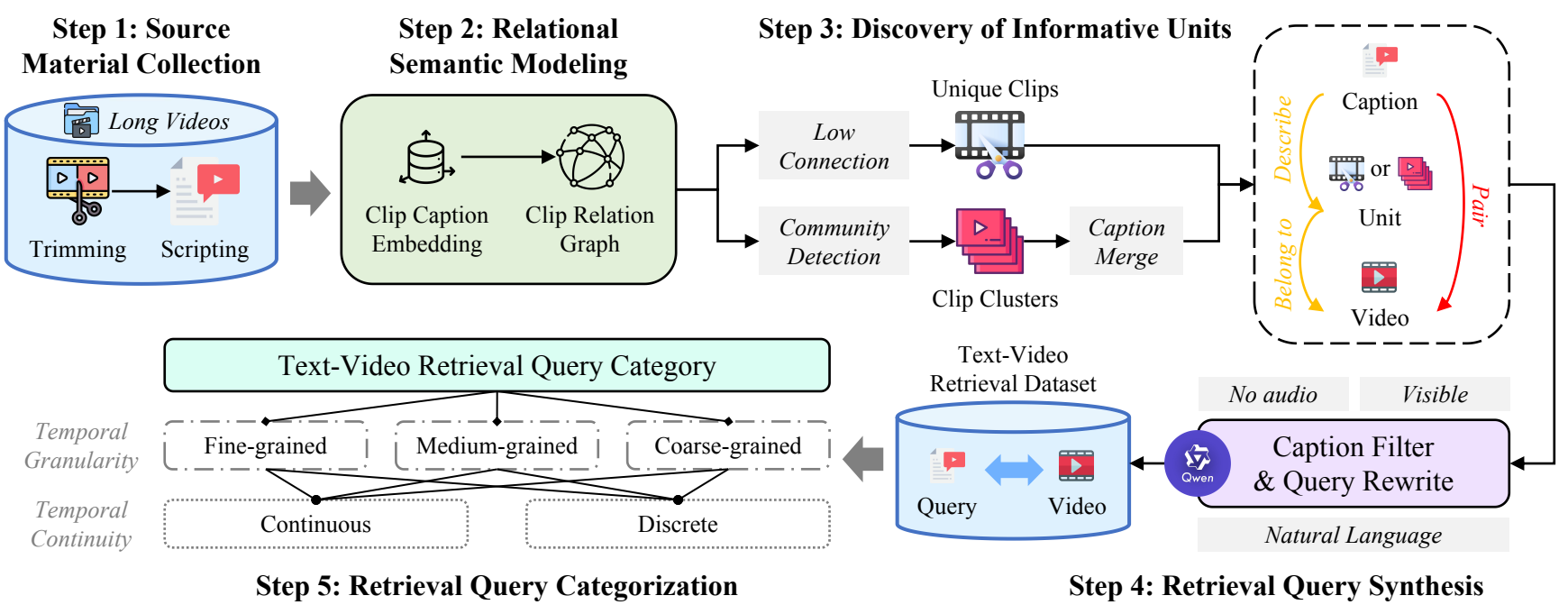

Step 1: Source Material Collection
Long Videos
Trimming
Scripting
Step 2: Relational Semantic Modeling
Clip Caption Embedding
Clip Relation Graph
Step 3: Discovery of Informative Units
Low Connection
Unique Clips
Community Detection
Clip Clusters
Caption Merge
Caption
or
Unit
Video
Describe
Belong to
Pair
Text-Video Retrieval Dataset
Query
Video
No audio
Visible
Qwen
Caption Filter & Query Rewrite
Natural Language
Step 4: Retrieval Query Synthesis
Text-Video Retrieval Query Category
Temporal Granularity
Fine-grained
Medium-grained
Coarse-grained
Temporal Continuity
Continuous
Discrete
Step 5: Retrieval Query Categorization